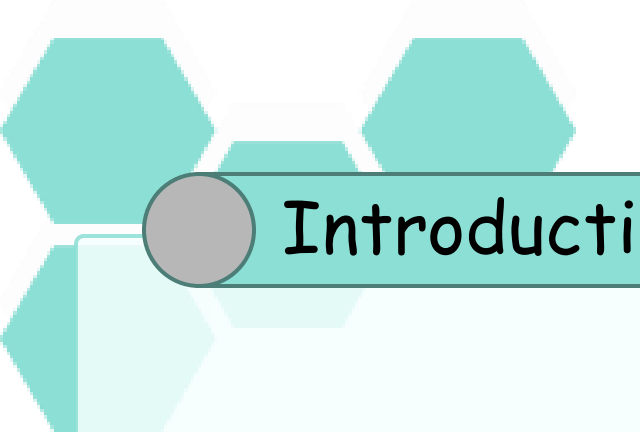


PEST and DISEASE MANAGEMENT in NATURAL FARMING



Ishwar Poojar
Natural Farming
MANAGE



Introduction

Natural Farming

Natural Farming is a holistic farming system which helps in improving soil regeneration, water and air quality, enhance ecosystem biodiversity, produce nutrient-rich food and store carbon to help mitigate the effects of [climate change](#). The farmers in local groups /clusters will be mobilized to work in harmony with nature, while also maintaining and improving economic viability.



Introduction

Techniques and practices of Natural Farming

- Minimize the physical, biological, and chemical disturbance of the soil
- Keep the soil covered with vegetation or natural material
- Bio diversity
- Integrate animals into the farm as much as possible
- Use of on farm inputs/ bio formulations



Pest and Disease management

The major aim for the natural farmer is to create conditions, which keep a plant healthy and less vulnerable to pest and disease (biotic and abiotic stress) infestation. The way to healthy plant is a healthy soil. A diverse cropping system will not allow an insect to build up to the level where it causes economic damage.



Types of damage



PEST



DISEASE



NON-INSECT
PESTS



DISORDER

Pest identification

Type of Pests

- Regular pest- Frequently- Rice stem borer, Pod borer
- Occasional - Case worm in rice, Mango stem borer
- Seasonal Pests- Red Hairy Caterpillar, Cotton pink bollworm, Mango Hopp
- Persistent pests- Round the year- Thrips, Mealy bug, cotton ball worm

Based on feeding mechanism:

- Sucking pests (these also transmit viral diseases)
- Borers/Chewing (leaf eaters)



Aphids



Bugs and Beetle



Mites



Caterpillars



Thrips & Flies



Grubs and Beetles

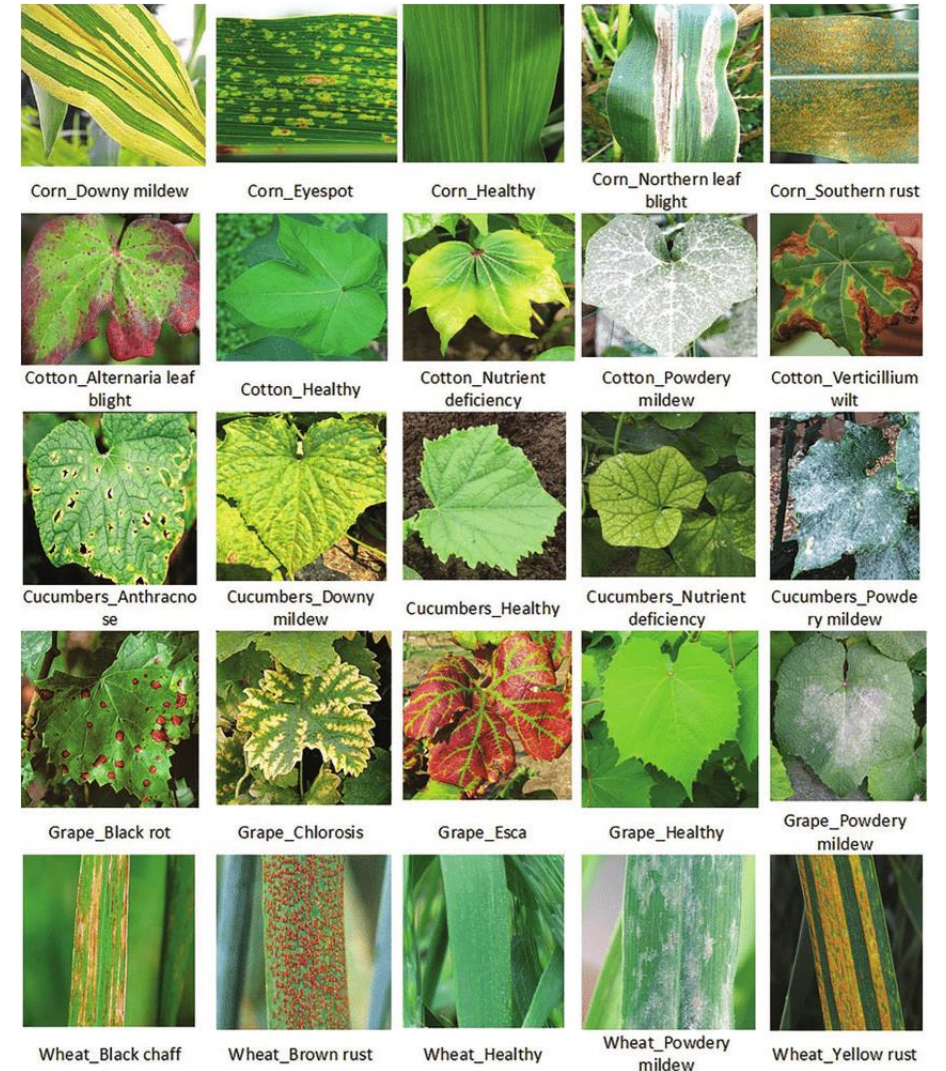


Disease identification

Type of Diseases

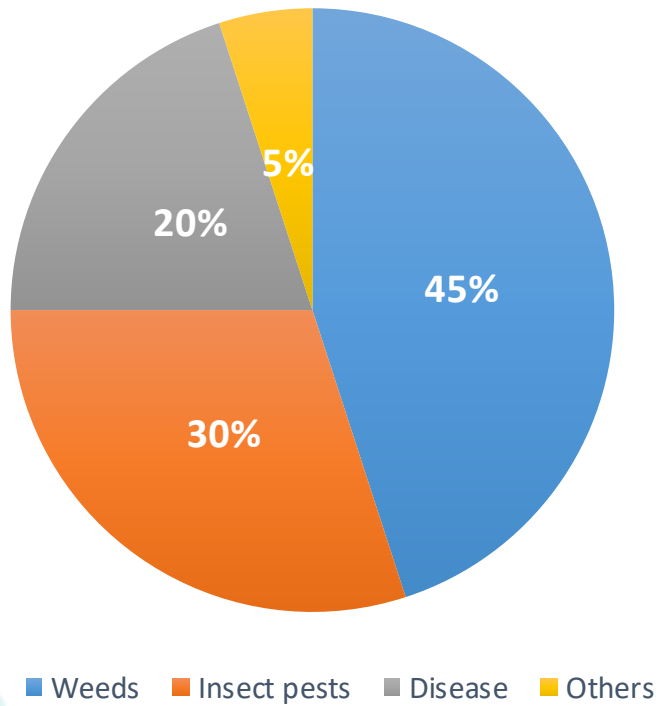
- **Bacteria**- Black rot - *Xanthomonas campestris*
Bacterial canker - *Clavibacter michiganensis*
- **Viruses**- Tomato spotted wilt
Cotton leaf curl
- **Fungi**- White blister/White rust (*Albugo candida*)
Sclerotium rots (*Sclerotium rolfsii* and *S. cepivorum*)
- **Nematodes**- Aphelenchoides, a disease of rice, is provoked by *Aphelenchoides besseyi*.
Tylenchulus semipenetrans cause serious citrus diseases

In Natural farming the outbreak of disease and spread of disease is controlled by periodical application of Beejamruta Dashaparni, Astras and Jeevamrutha.

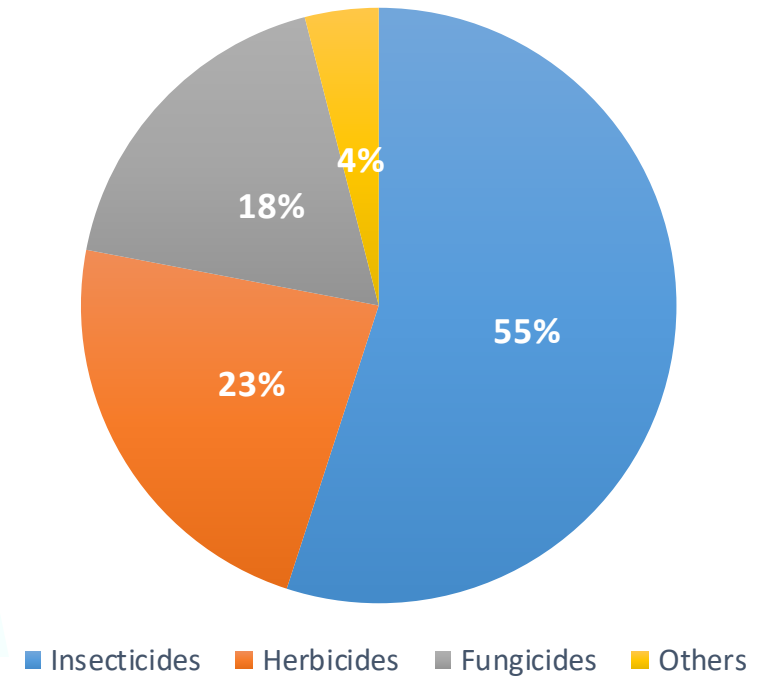


Crop losses

Crop loss by different categories



Chemical usage

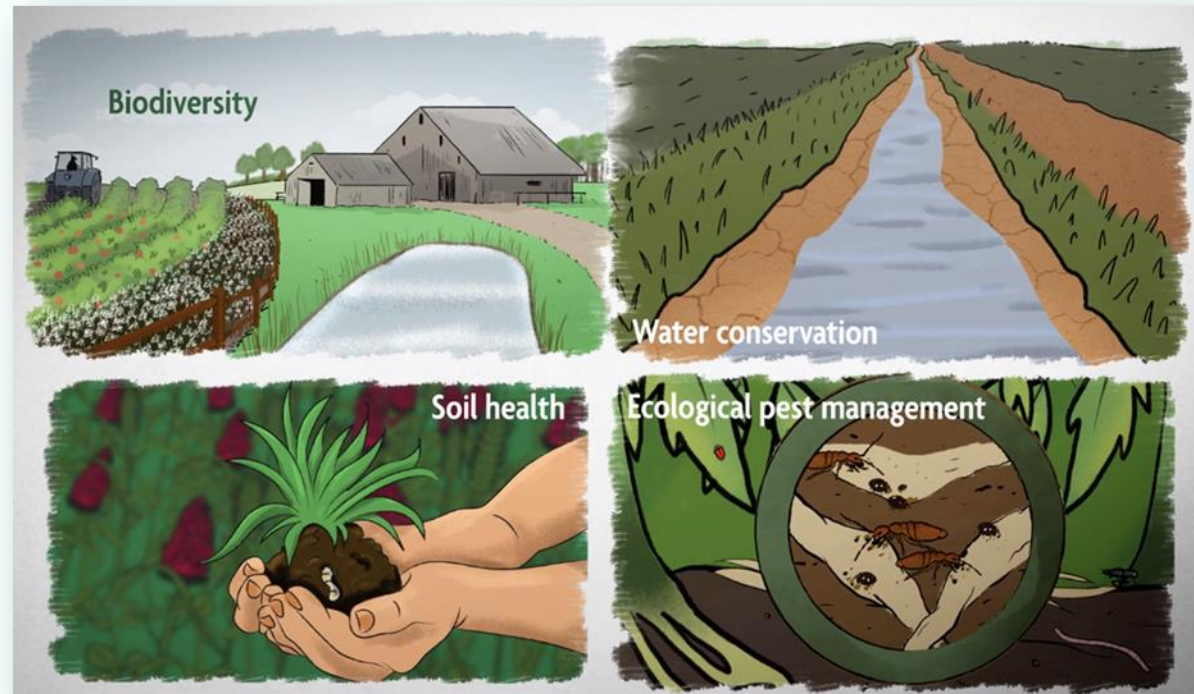


Causes for Outbreak of Pests/Disease

1. Destruction of forest or bringing forest area under cultivation
2. Indiscriminate use of pesticides leads to destruction of natural enemies, pest resistance, pest resurgence
3. Intensive cultivation
4. Introduction of new crops and varieties (many high yielding varieties are more susceptible to insects)
5. Improved agronomic practices (higher 'N', close, spacing, weed control etc. improved crop growth and reduced competition for food to the insects)
6. Introduction of new pest in a new area
7. Accidental introduction of foreign pests
8. Large scale storage of food grains (outbreak of stored product pests, rat problem)

Baseline for Pest & Disease Management in NF

- A Well Managed Ecosystem
- Local Resistant Crop Varieties
- Avoid Monocropping
- Health condition of plants depends on Soil Health
- Favorable Climatic Conditions
- Biodiversity



Pest & Disease Management

1. Preventive Measures



2. Monitoring



3. Curative Methods



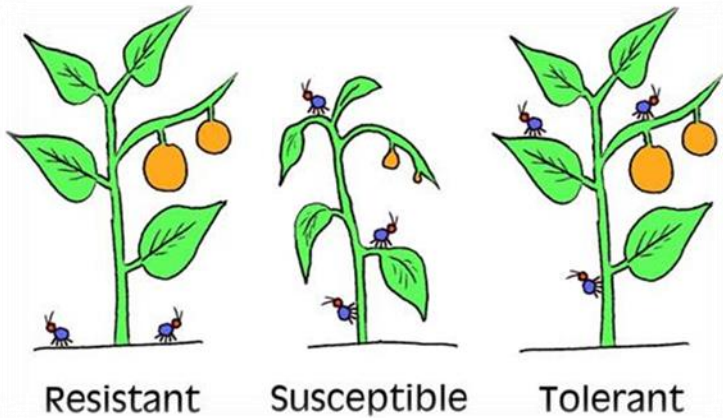
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Preventive Measures

- A. Selection of varieties well adapted to local environment
- B. Selection of Safe Seed/Planting Material
- C. Adopting Mix Cropping System/Diversified cropping
- D. Crop rotation
- E. Conservation & Prevention of Natural Enemies
- F. Choosing optimal planting time & Sufficient Spacing between crops
- G. Remove infected plant parts

A.

Selection of varieties well adapted to local environment



There is most variety in the traditional crops grown by farmers

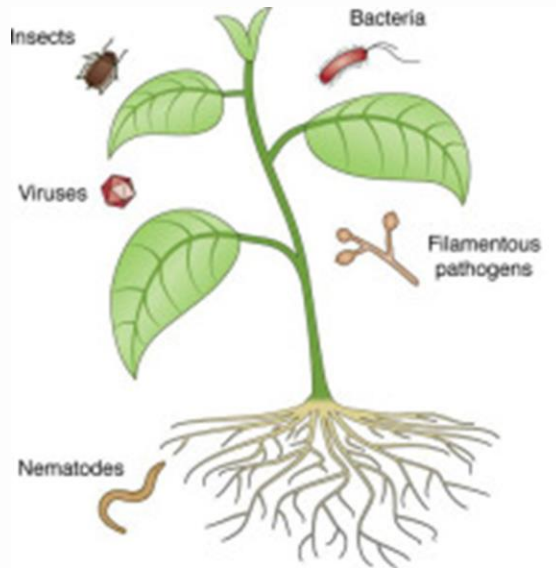


Crops which have been bred by modern breeding methods tend to be very similar and if one plant is susceptible to a disease, all the other plants are as well



Although some new modern varieties may be very resistant to specific pests and diseases they are often less suited to the local climate and soil conditions than traditional varieties

Plant immunity



The creatures that we call pests and the organisms that cause disease only become 'pest and diseases' when their activities start to damage crops and affect yields



The aim of natural control is to **restore a balance between pest and predator** and to keep pests and diseases down to an acceptable level



The aim is not to eradicate them altogether, as they also have a role to play in the natural system

B.

Selection of Safe Seed/Planting Material

- **Local adaptation:**
- **Disease-free material:**
- **Genetic purity:**
- **Heirloom Varieties:**
- **Treated with Natural Concoctions & Decoctions:**



C.

Adopting Mix Cropping System/Diversified cropping



Companion cropping: Growing two or more crops on a field at the same time decreases the risk of pests and disease infestations.

Companion planting



growing onions with carrots to help deter
carrotfly



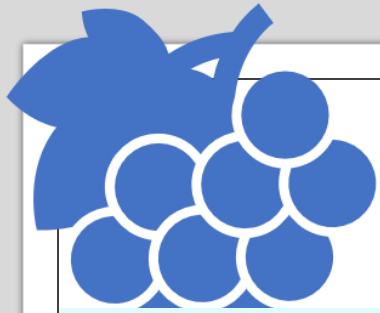
Companion planting means growing certain plants to protect other plants from pests or diseases



This may be because the pest is deterred by the companion plant, or because it is attracted to the companion plant rather than the crop



Many pests avoid garlic, so this can be used very effectively for companion planting with most crops

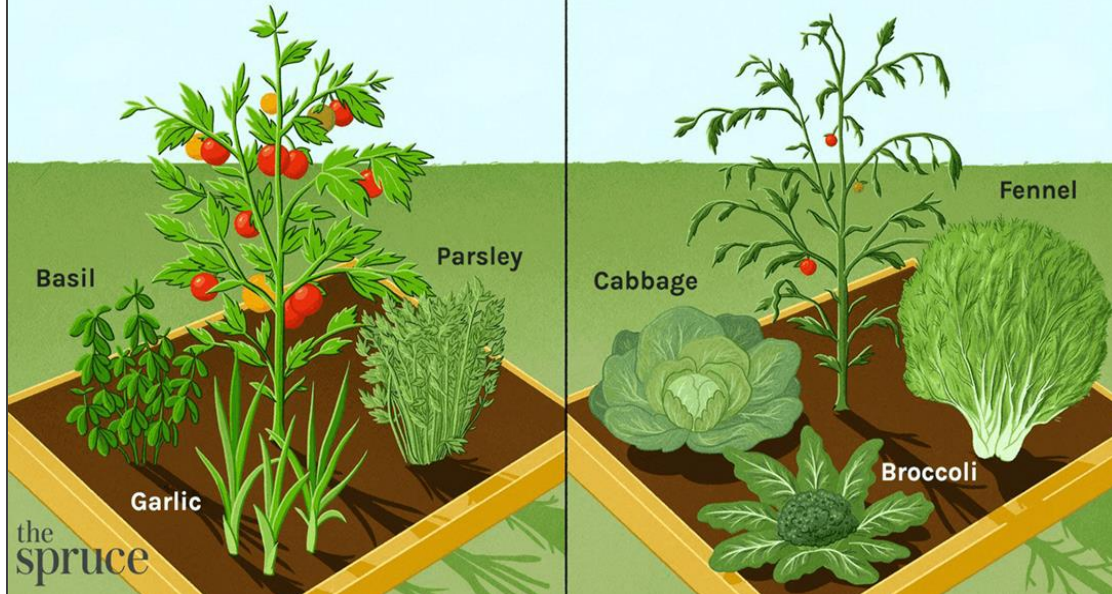


Companion planting

Best and Worst Companion Plants for Tomatoes

These plants improve tomato plant growth and flavor, and repel bugs.

These plants stunt or inhibit tomato plant growth.



Companion Planting Guide



Trap crop

A trap crop is a plant that attracts agricultural pests, usually insects, away from nearby target crops.

Main crop	Trap crop		Pest managed	
Cotton Groundnut Tobacco	Castor		Tobacco caterpillar	
Tomato Groundnut Cotton	Marigold		American Bollworm	
Cabbage Cauliflower Radish	Mustard		Diamondback moth	
Cotton	Onion		Leaf thrips	



Barriers

- Barriers are physical structures put in place to prevent a pest from reaching a plant
- They keep pests away from a plant but do not kill them
- Here are some examples that you can adapt, depending on the resources available to you





Specific plants to attract beneficial insects



There are many plants that can be grown to attract natural predators and parasites which will help to keep down pests and diseases



Hoverflies, whose larvae feed on greenfly are attracted to the flowers of herbs and vegetables such as fennel, celery, dill, carrots and parsnips



Umbellifers will also provide food to various parasitic wasps whose young live on aphids and some caterpillars



Specific plants to attract beneficial insects



			
Lacewing bug	<i>Achillea filipendulina</i> Fern-leaf yarrow	<i>Anethum graveolens</i> Dill	<i>Anthemis tinctoria</i> Golden marguerite
			
Praying mantis	<i>Cosmos bipinnatus</i> Cosmos white	<i>Helianthus maximilianii</i> Prairie sunflower	<i>Callirhoe involucrata</i> Purple poppy mallow
			
Wasp	<i>Coriandrum sativum</i> Coriander	<i>Taraxacum officinale</i> Dandelion	<i>Tanacetum vulgare</i> Tansy
			
Lady bug	Carpet bugleweed	Hairy vetch	Marrigold

Similarly to companion planting, which seeks to deter pests from the main crop, attractant plants can be grown to attract predatory insects



Areas of natural habitat



Bushes and trees are a home for many useful insects and birds



Areas of natural habitat can be left around the edges of fields where crops are grown

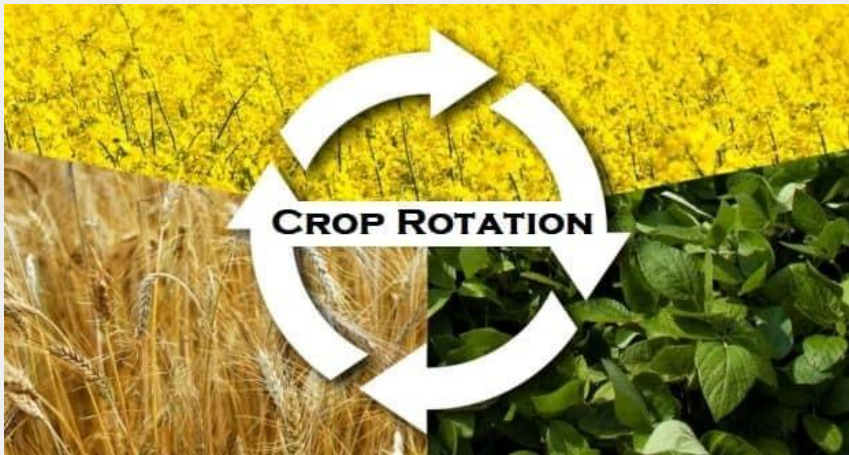


If these areas are destroyed then there is likely to be an imbalance between the populations of predator and pest

D. Crop rotation



Crop rotation
helps to control
pests and diseases



With crops such as brassicas and onions which are usually grown in a vegetable garden the whole year round, the populations of certain pests and diseases can keep increasing because there is always a suitable host plant for them



Breaking the cycle can help to solve the problem



This can be done through rotation within the vegetable garden

Maintain a balanced and diverse crop rotation: This increases diversity of crops over time and limits the build-up of pest and diseases

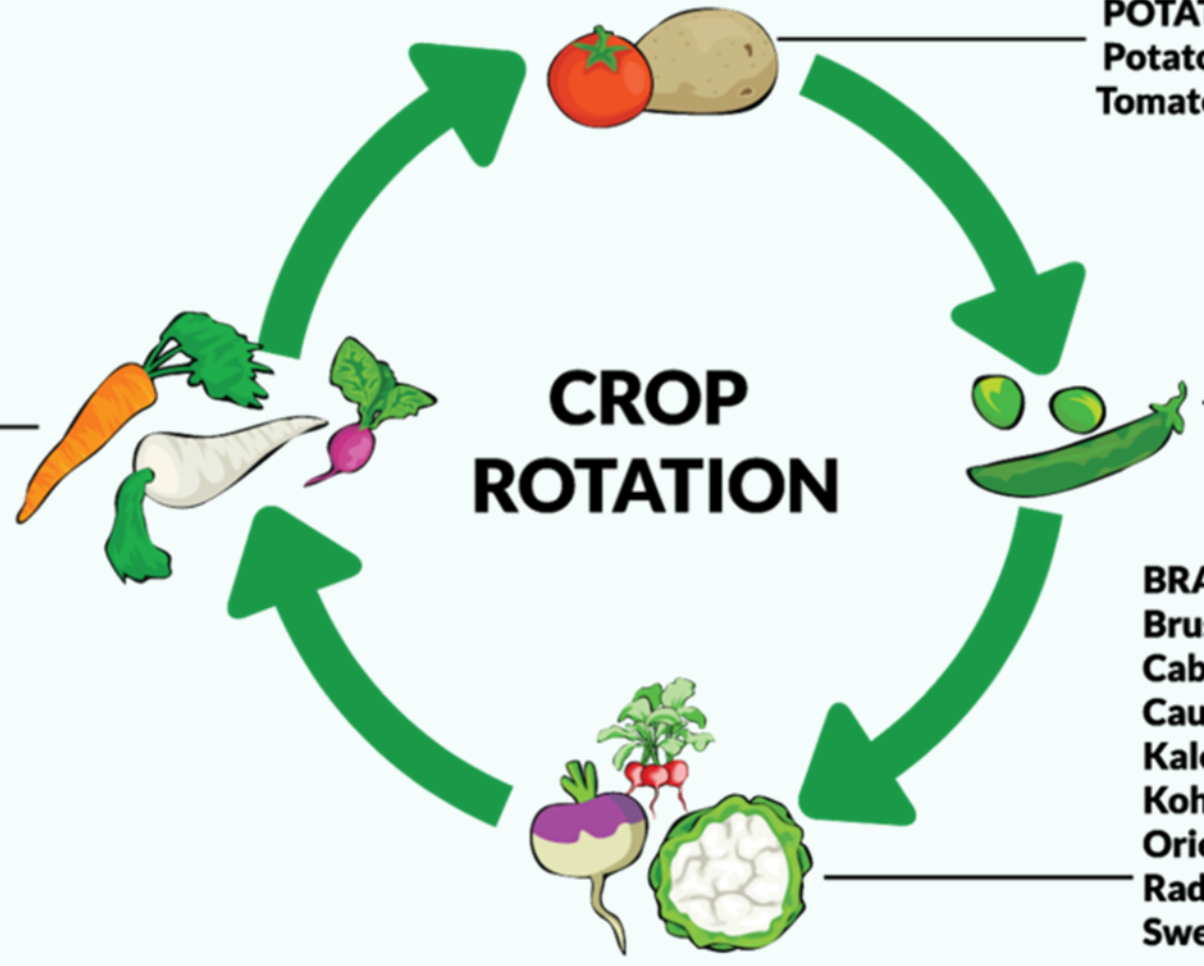
CROP ROTATION

POTATO FAMILY:
Potato
Tomato

LEGUMES:
Peas
Broad beans
French beans
Runner beans etc.

BRASSICAS:
Brussels sprouts
Cabbage
Cauliflower
Kale
Kohl-rabi
Oriental greens
Radish
Swede and Turnips

ROOTS:
Alliums
Beetroot
Carrot
Celeriac
Celery
Florence fennel
Parsley
Parsnip
and all other root crops



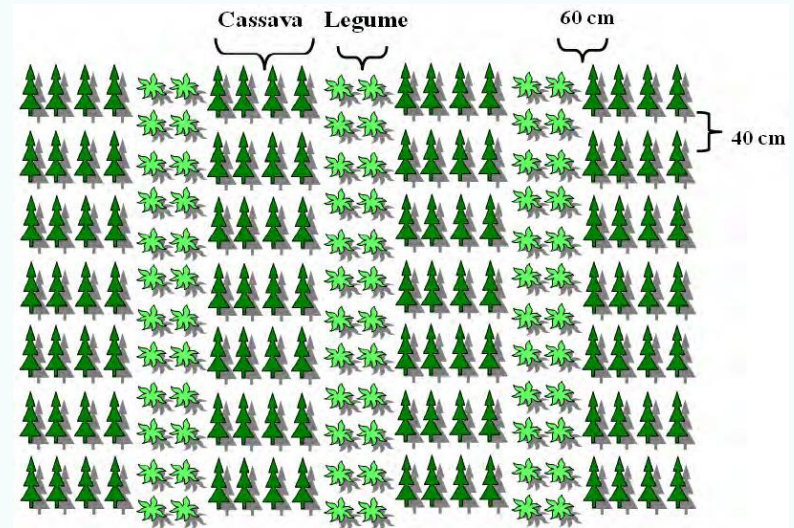
Inter cropping

Growing two or more crop species at the same time in the same field is termed intercropping

The main purpose of intercropping is to allow for more important biological and crop interactions.

Ex: cassava brown streak disease was significantly suppressed by intercropping with a legume

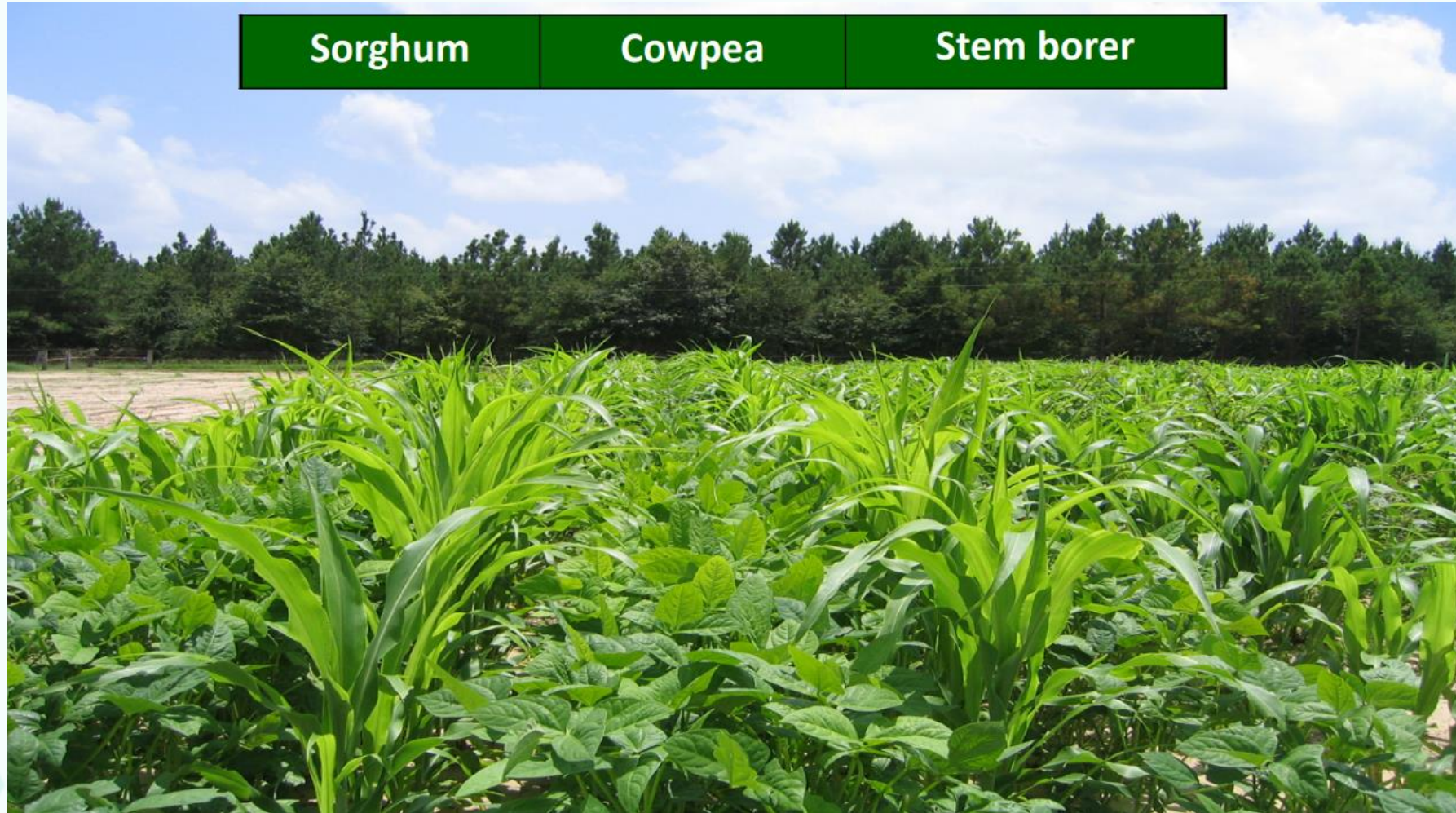
- Disease pressure was reduced in the intercropping field
- Check the movement of dispersal of the pathogen



Inter cropping

S. No.	Crop		Pest reduced	Reference
	Main crop	Intercrop		
1.	Sorghum	Red gram	Earhead bug	Raheja (1973)
2.	Sorghum	Cowpea	<i>Chilo partellus</i>	Balasubramainian (2000)
3.	Pigeon pea	Sorghum	<i>Empoasca kerri</i>	Sekhar et al (1997)
4.	Green gram	Sorghum	<i>E. kerri</i>	Sekhar et al (1997)
5.	Ground nut	Sorghum	<i>E. kerri</i>	Sekhar et al (1997)
6.	Pigeon pea	Sorghum	<i>H. armigera</i>	Mohammed and Rao (1998)
7.	Chickpea	Wheat/ mustard/ Safflower	<i>H. armigera</i>	Das (1998)
8.	Sugarcane	Green gram / Black gram	Early shoot borer	Rajendran <i>et al.</i> (1998)

Inter cropping

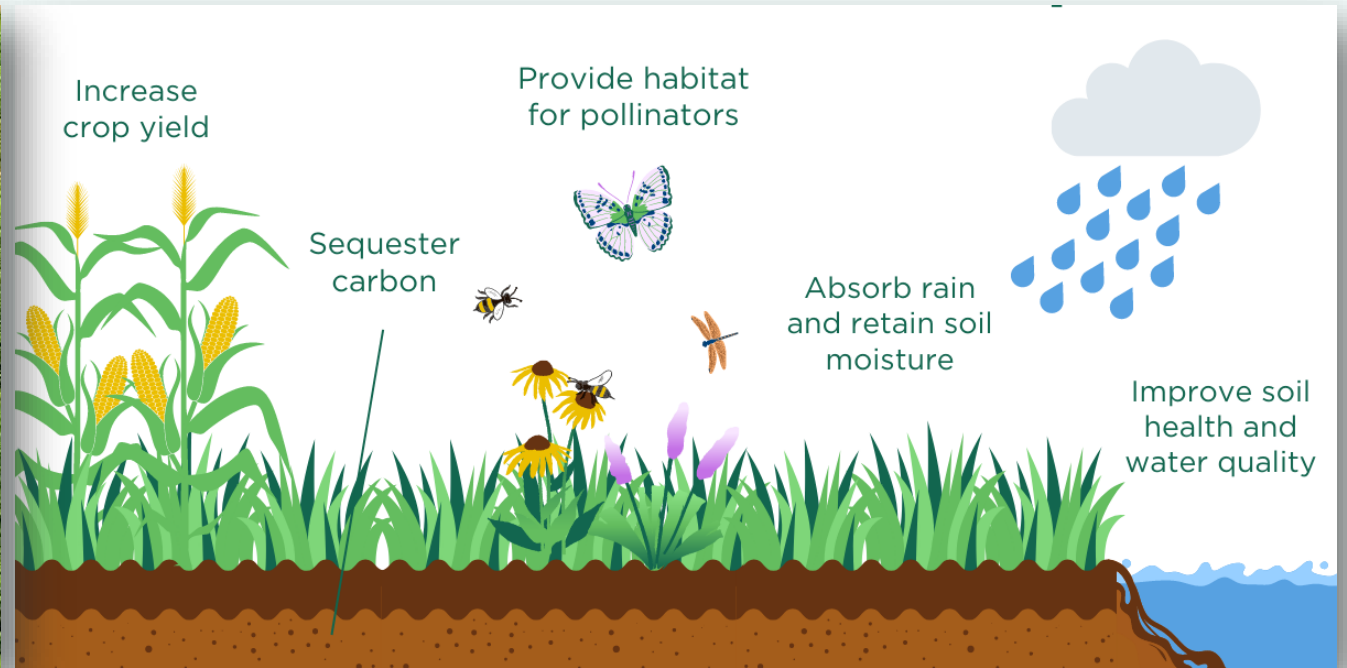


Cover cropping

A cover crop is defined as any living ground cover that is planted with or after the main crop, and usually killed before the next crop is planted

The use of these systems had been greatly reduced with the introduction of herbicides and synthetic fertilizers

- increase organic matter and reduce weeds, pests and soil erosion
- prevent the movement of disease spores onto the crop



Conservation & Prevention of Natural Enemies



Predators capture and eat other organisms such as insects or mites



Parasitoids are insects that parasitize other insects



Pollinators include honeybees, other wild bees, butterflies, moths and other insects that visit flowers to feed on nectar and pollen

Cow pea and Lab-lab as border crops to conserve and increase the activity of natural enemies





Timely sowing



F

Choosing optimal planting time



It is often the young of many pests, rather than the adults, that cause damage to crops



Problems can be avoided by delaying sowing until the egg laying period of a pest is over, or by protecting the plants during this period



It is therefore important to know the life cycle of pests, so that timely sowing can be carried out

Role of planting dates on pest population

No.	Host plant	Insect	Response	Reference
1.	Rice	Leaf folder	Early planted rice (up to 3 rd week of June) suppressed population	Dhaliwal <i>et al.</i> (1988)
2.		BPH	Planting during end of July in Kharif and early in Rabi escapes attack	Krishnaiah <i>et al.</i> (1986)
3.		Gall midge	Lowest incidence if planted in Aug or Oct	Uthamasamy and Karuppuchamy (1986)
4.	Sorghum	Shoot fly	Advancing sowing date (Sept - Oct) decreased incidence	Kotikal and Panchbavi (1991)
5.	Cotton	Leafhopper	Higher incidence in late sown crop	Dhawan <i>et al.</i> (1990)
6.	Chickpea	<i>H. armigera</i>	For every 10 day delay in sowing 4.02% increase in pod damage	Devendra Prasad <i>et al.</i> (1989)
7.	Tomato	Whitefly	Incidence less if planted within July - Nov.	Saikia and Muniappa (1989)
8.	Chillies	Thrips	Late planted crop severely affected by thrips and leaf curl virus	Bagle (1992)

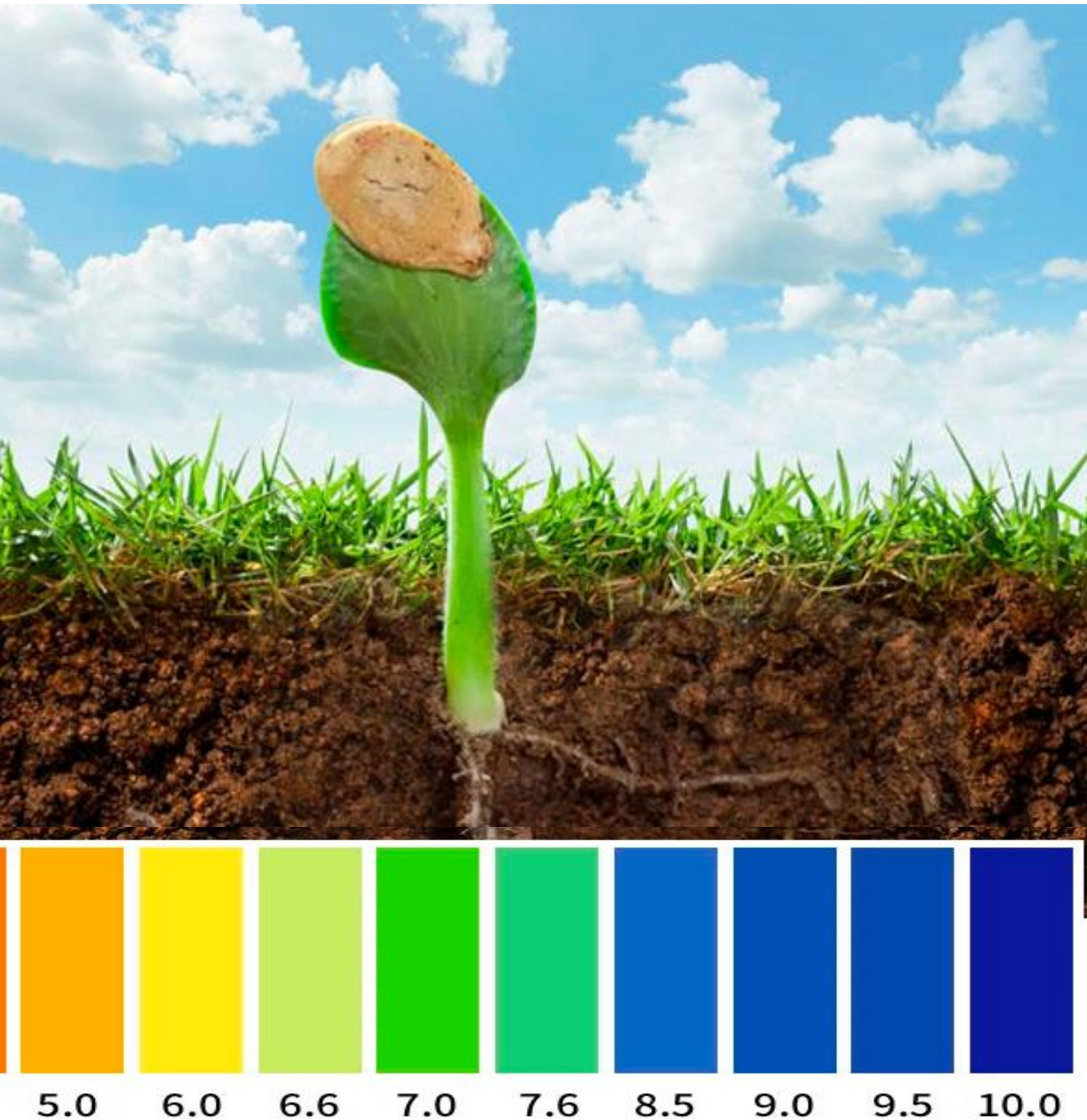
Effect of plant density on pest population

No.	Crop	Spacing/ density	Insect	Response	Reference
1.	Rice	Dense planting	Leaf folder, BPH	High incidence	Kushwaha and Sharma (1981) Kalode and Krishnaiah 1991)
2.	Chickpea	Dense plant population	<i>H. armigera</i>	High incidence	Yadav (1987)
3.		Dense plant population	<i>Aphis craccivora</i>	High incidence	Lal <i>et al</i> (1989)
4.	Sugarcane	Dense seed rate	Top shoot borer	Low incidence	Singla and Duhra, 1990
5.		Dense seed rate	Early shoot borer	High incidence	

Soil tillage



- Many pests spend part of their lives as larvae or pupae in the soil
- Ploughing or digging when the soil is dry can reveal the pest and they will dry out and die in the sun, or they can be picked off the ground by hand or birds or other predators
- Ploughing and disturbing the soil should be carefully considered against the harmful effects it may have such as destroying the structure of the soil and causing soil erosion



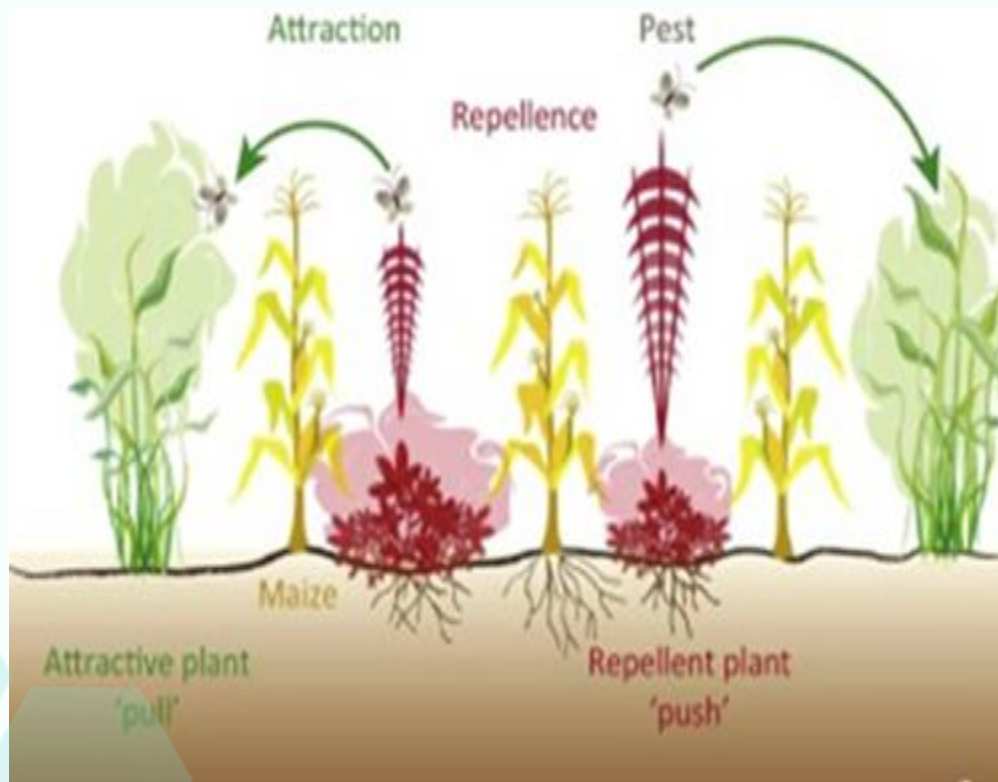
Soil pH

- The pH of a soil can affect some diseases
- A layer of grass mowings added to the bottom of the potato trenches at planting time will make the soil more acid and reduce the disease
- Clubroot is less severe in alkaline conditions therefore liming the soil to make it more alkaline can reduce the problem

- A. Regular Monitoring
- B. Must be aware about specific pest & its damage in the region
- C. Most crop pests belong to the insects, mites and nematodes
- D. Insect damage can be categorized by
 - biting and chewing (ex: caterpillars, weevils),
 - piercing and sucking (ex: aphids, psyllids) and
 - boring (ex: borer, leaf miner)
- E. Some are slow moving (ex: caterpillars),
 - fast moving (ex: fruit flies),
 - hidden (ex: stem borer),
 - easy to observe (ex: caterpillars, weevils)
- F. Pest/Disease damage is often Specie - specific

3 Curative Methods

A. Promoting & Managing Natural Enemies

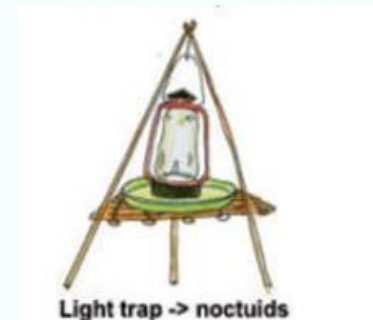


- Natural Enemies of Pests are other organisms which kill pest
- Minimize the application of natural pesticides
- Establish a diverse cropping system
- Include host plants providing food or shelter for natural enemies

Curative Methods

B. Mechanical Control

- Light Traps - Moths such as Armyworms, cutworms
- Colour & Water Traps - Adult Thrips
- Yellow Sticky Traps - White flies, Aphids
- Bait Traps - Flies
- Fruit Bagging - Fruit flies



Light traps

The use of light traps are traditional methods, used against many adult lepidopteran pests
Ex: fruit moths



Colour/water traps

The use of colour traps are traditional methods, used against many sucking pests

Ex: Thrips



Bait traps

The use of baits and traps are traditional methods which have become neglected because of the increasing use of chemical pesticides

Ex: *Spodoptera* pest



Mechanical methods of pest management

- Installation of bird perches and light traps are most effective mechanical methods of pest control
- 'T' type of bird perches with 5-6 feet height



G. Remove infected plant parts

Good hygiene

- If infected plant material, live or dead, is left lying around, pests and diseases can be passed on to future crops
- Debris should be cleared up and disposed of
- The composting process will kill some pests and diseases and produce compost which is a good soil improver and fertilizer



Mechanical methods

Mechanical destruction

- Hand picking of caterpillars
- Flame thrower - moringa hairy caterpillar
- Hooking of rhinoceros beetle adult with iron hook
- Sieving and winnowing for stored product insect control
- Shaking plants - to dislodge caseworm in rice, June beetles from neem trees



Mechanical methods ...

Mechanical exclusion

- Wrapping fruits - pomegranate fruit borer
- Red earth coating - red gram - pulse beetle
- Banding with grease - mango mealy bug
- Trenching - for larvae of red hairy caterpillar
- Tin barrier - coconut tree trunk - rat damage
- Rat proof structure in storage godowns



C. Plant Based Concoctions & Decoctions

Beejamrit

It is a mixture of cow dung, cow urine, water, lime and a handful of soil.

Ingredients Used

- Water - 20 liter
- Cow Dung - 5 kg
- Cow Urine - 5 liter
- Lime - 50 gram
- Soil - Handful

Preparation

- Take 5 kg desi Cow dung and wrap it in a cotton cloth
- Take 20 liter water in bucket and dip the above 5 kg cow dung wrapped in cloth into it
- Leave it for 12 to 16 hours so that the cow dung extract may come into the water
- Take 50gm lime in another container having 1 liter water
- Now mix the above two preparations and into it add 50gm soil (Rhizospheric soil from tree or from bund)
- Add 5 liter of cow urine into it and leave the solution prepared for 8-12 hours
- Now the Beejamrit is ready for seed treatment



C. Plant Based Concoctions & Decoctions

Beejamrit

Application methods:

Seed treatment: 20 liters of Beejamrit is sufficient for treatment of 100 kg seeds. Apply solution to the seeds and mix it properly, dry the seeds in shade and use for sowing

Seedling treatment: If vegetative part used for cultivation then dip vegetative part like seedling, rhizome, stem, root, leaf cuttings, tuber etc. for 5 minutes.

Benefits of Beejamruta

- Kills all disease-causing spores of seed borne diseases
- Kills insect eggs associated with seeds
- Induce germination, terminates hibernation
- Uniform germination
- Healthy crop



Neemastra

Neemastra is a natural pesticide. It is a very effective insecticide for leaf miners and sap sucking insects (By using it, enemy insects and moths do not come out in the field)

Ingredients Used

- Neem Leaves - 10 Kg
- Cow Urine - 10 Litre
- Cow Dung - 2 Kg
- Water - 200 Litre



Neemastra

Preparation

- Take 10 kg of crushed neem leaves/neem dried fruits
- Add this crushed neem leaves in 200 liters of water
- Put 10 liters of cow urine in it and mix two kg of cow dung.
- Stir it with wood and keep it covered for 48 hours. Keep in shade places
- After 48 hours filter the solution with a cloth.
- Now solution is ready for spray



How to Use?

Spray 200 liter per acre

Uses:

For the management of sap sucking insects and small caterpillars.



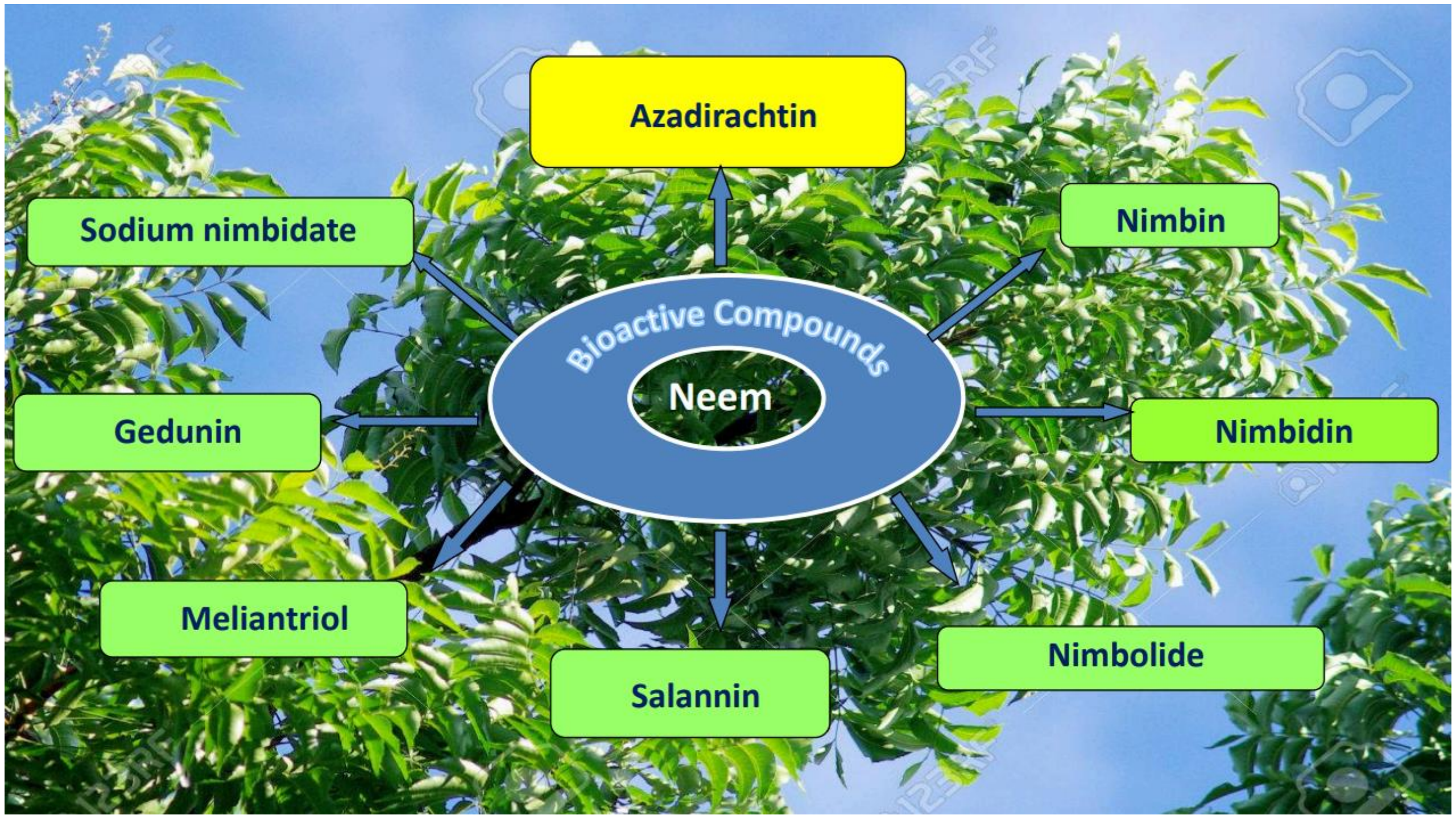
Plant Origin Natural Pesticides

- Derivatives from plants are specific to target pest, eco-friendly and less harmful to beneficial insects
- More than 2400 plant species - screened - pesticidal properties - 240 - promising
- Includes - antifeedants, sterilants, feeding & oviposition deterrents, growth regulators, ovicidal & insecticidal activities

Organic compounds of plant origin:

- Pyrethrum - Pyrethroids
- Rotenone - Rotenoids
- Neem - Azadirachtin





Azadirachtin

Nimbin

Nimbidin

Nimbolide

Salannin

Meliantriol

Gedunin

Sodium nimbidate

Neemastra

How to Use?

Spray 200 liter per acre

Uses:

For the management of sap sucking insects and small caterpillars.



Bramhastra

It is a natural insecticide prepared from leaves which contain specific alkalis (alkaloids) to repel insects. It controls all sucking insects and hidden insects present in pods and fruits.

Ingredients Used

- **Neem Leaves** - 5 Kg
- **Pogemia Leaves** - 2 Kg
- **Custard Apple Leaves** - 2 Kg
- **Datura Leaves** - 2 Kg
- **Guava Leaves** - 2 Kg
- **Lantana** - 2 Kg
- **Bitter gourd leaves** - 2 kg
- **Castor leaves** - 2kg
- **Parthenium leaves** - 2kg
- **Pomegranate leaves** - 2 kg
- **Vitex spp leaves** - 2kg
- **Cow Urine** - 20 liter



Plant Materials required for Bramastra Preparation

Lantana camara



Datura leaves



Neem leaves



Pomegranate leaves



Guava leaves



Custard apple leaves



Castor leaves



Bitter gourd leaves



Parthenium leaves



Pongamia
leaves
(ಹೊಂಗೆ ಎಲೆ)



Vitex spp
leaves



Bramhastra



Preparation:

- Take 20 liters of cow urine
- Add any 5 types crushed leaves contents and mix thoroughly clockwise with the help of a stick, close it with a lid and boil the contents
- After 48hrs, filter the solution with a thin cloth After 3-4 boils, cool it
- Stir the solution morning and evening for a minute for 3 days
- Solution is ready to spray on the crop

How to Use?

- 2-3% Spray with water or 6 liter in 200 liter of water for acre

Uses:

- For the control of sucking insects and pod/fruit borer.

Agniastra

Ingredients Used

- Neem Leaves - 5 Kg
- Green Chilli - 0.5 Kg
- Garlic - 0.5 Kg
- Tobacco leaves - 1 kg
- Cow Urine - 20 Litre

Green chilli



Garlic



Tobacco leaves



Agniastra

Preparation

- Take 20 liters of cow urine
- Add 05 kg of crushed green leaves of neem
- Add 0.5 kg crushed Green Chilli
- Add 0.5 kg crushed Garlic and 1 kg tobacco leaves
- Now dissolve all this mixture in cow urine and boil it
- After 3-4 boils, take it down from the fire
- Let it cool for 48 hours and then filter the solution with a cloth
- Now Solution is ready to spray on the crop

How to Use?

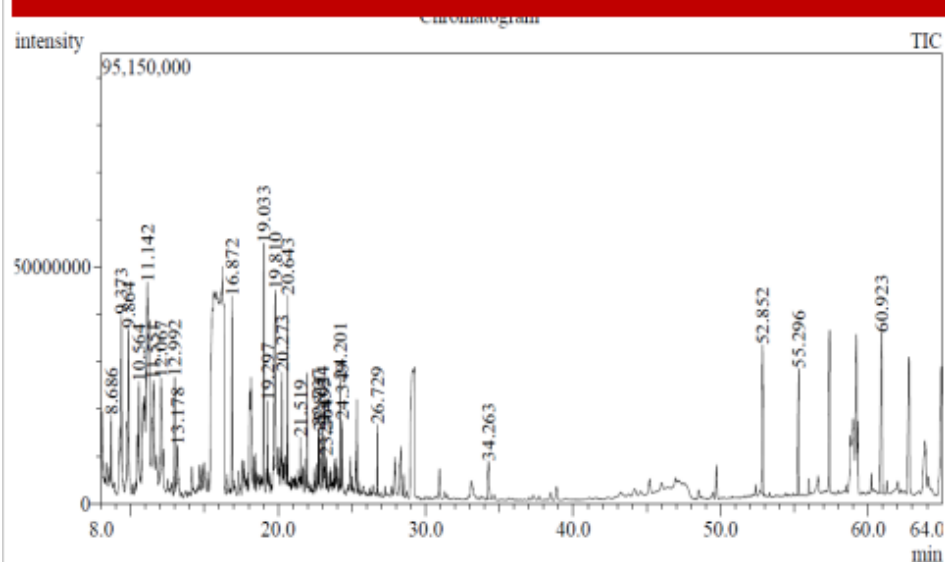
2-3% Spray with water or 6 liter in 200 liter of water for acre

Uses

For insects living in tree trunks or stalks, all types of large bollworms and caterpillars



Analysis of bioactive compounds in green chilli by GCMS



Peak Report TIC

Peak#	R.Time	Area	Area%	Name
1	8.686	38188758	1.84	Benzene, 1,2-dimethyl
2	9.373	92768142	4.46	Pentane, 3-ethyl-3-methyl-
3	9.864	94848036	4.56	Valeric acid, 2-ethoxyethyl ester
4	10.564	62395562	3.00	Pentane, 3-ethyl-2,4-dimethyl
5	11.142	88655502	4.27	Glutaric acid, hexyl 4-methylbenzyl ester
6	11.551	55233620	2.66	Carbolic acid
7	12.067	15536928	5.56	Mesitylene
8	12.992	96933095	4.67	DECANE
9	13.178	34676477	1.67	BENZENE, 1,3,5-TRIMETHYL
10	16.872	21074672	5.83	Undecane
11	19.033	33364927	6.42	TETRADECANE
12	19.297	31171249	1.50	Undecane, 2,5-dimethyl-
13	19.810	15582682	10.38	Benzene, 1-ethoxy-4-methyl-
14	20.273	37417542	1.80	Pentadecane, 2,6,10,14-tetramethyl-
15	20.643	79854654	3.84	Tetradecane
16	21.519	16906896	0.81	Tetradecane, 2,6,10-trimethyl-
17	22.737	21902977	1.05	Eicosane, 10-methyl-
18	22.811	14569785	0.70	2-CYCLOHEXEN-1-ONE, 4-BUTYLIDENE-3-METHYL-5-PROP
19	23.024	23035209	1.11	Bicyclo[3.1.1]hept-2-ene, 2,6-dimethyl-6-(4-methyl-3-pentenyl)-
20	23.135	15743204	0.76	PENTADECANE
21	23.264	12546351	0.60	2-Methoxy-4-methyl-1-pentylbenzene
22	24.201	42829544	2.06	Heptadecane
23	24.349	31155335	1.50	(R)-5-(HYDROXYETHYL)-6-METHYLHEPT-6-EN-2-ONE
24	26.729	33312358	1.60	Heneicosane
25	34.263	53058231	2.55	2-HEXADECEN-1-OL, 3,7,11,15-TETRAMETHYL
26	52.852	70817215	8.22	1,2-BENZENEDICARBOXYLIC ACID, DITRIDECYL ESTER
27	55.296	29990322	6.26	Hexatriacontane

Bioactive compounds present in ginger, garlic and green chilli extract

Chemical formula	Name of the bioactive compound	Uses
$C_{24}H_{50}$	Tetracosane	Antifungal
$C_{36}H_{74}$	Hexatriacontane	Antimicrobial
$C_{15}H_{24}$	Beta sesquiphellandrene	Antimicrobial
$C_{10}H_{16}O$	2,6-octadienal 3,7-dimethyl- (z)	Antimicrobial
$C_6H_8S_2$	3-vinyl-1,2-dithiacyclohex-4-ene	Antibacterial
$C_6H_{10}S_2$	Diallyl disulfide	Antimicrobial
$C_6H_8S_2$	1,2-dithin, 3-ethenyl-3,4-dihydro	Antimicrobial
$C_6H_8S_2$	2-vinyl-4h-1,3-dithiin	Antioxidant
$C_{14}H_{30}$	Tetradecane	Antifeedant

**3G Extract 5 % - Sucking pests and defoliators
- Cotton & Vegetables**

Dashparni Extract

One stop concoction for various pesticides. This can be used instead of all the above astrams



Ingredients Used

- 200 lit Water
- 20 lit Desi Cow Urine
- 2 kg Desi Cow Dung
- 200 gm Turmeric Powder
- 500 gm Ginger Paste
- 10 gm Hing (Asafoetida) powder
- 1 kg Tobacco powder
- 1 kg Hot Green chilli Paste
- 500 gm Desi Garlic Paste

Ingredients Used

1. 2kg Neem leaves and twigs cut into smaller pieces
2. 2kg Karanj / Ponga leaves (Not paste)
3. 2kg Custard Apple Leaves (Not Paste)
4. 2kg Castor Leaves (Not Paste)
5. 2kg Dhatura leaves (Not Paste)
6. 2kg Bael Leaves (Not Paste)
7. 2kg Krishna Thulasi stem, leaves, seeds. cut into small pieces
8. 2kg Marigold root, branch, stem, flower, leaves. cut into small pieces
9. 2kg Calotropis leaves
10. 2kg *Nerium oleander* leaves
11. 2kg Mango leaves
12. 2kg Papaya leaves
13. 2kg Hibiscus leaves
14. 2kg Moringa leaves
15. 2kg Ber leaves
16. 2kg Babul tree (*Acacia nilotica*) with stem and leaves
17. 2kg Turmeric leaves cut into pieces
18. 2kg Ginger leaves cut into pieces
19. 2kg Coffee leaves cut into pieces
20. 2kg Bouganville leaves with flowers cut into pieces
21. 2kg *Casia tora* leaves and flowers
22. 2kg Pomegranate leaves



Note: First five leaves is compulsory and choose 5 types of leaves from remaining list as per availability



2kg Neem leaves



2kg Karanj / Ponga leaves



2kg Custard Apple Leaves



2kg Castor Leaves



2kg Datura leaves



2kg Bael Leaves



Krishna Thulasi stem



2kg Marigold



2kg Calotropis leaves



2kg Nerium oleander leaves



2kg Mango leaves



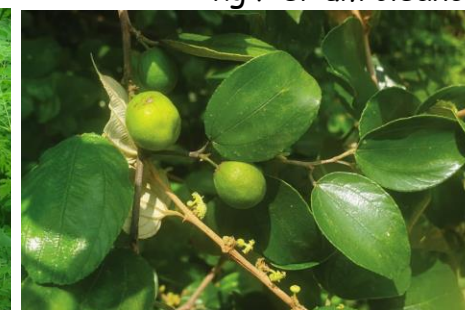
2kg Papaya leaves



2kg Hibiscus leaves



2kg Moringa leaves



2kg Ber leaves



2kg Babul tree



2kg Turmeric leaves



2kg Ginger leaves



2kg Coffee leaves



2kg Bouganville leaves



2kg Casia tora leaves



2kg Pomegranate leaves

Dashparni Extract

Preparation:

Step 1: Take 180 litres of water in a drum, add 20 litres of cow urine and 2 kg of cow dung. Mix it well and cover with the gunny bag and keep aside for 2 hours

Step 2: Add 500 gram of turmeric powder, 200 gram of ginger paste, 10 grams of Asafoetida into the mixture. Stir it well in the clockwise direction; cover with gunny bag and keep overnight

Step 3: Next morning, add 1 kg of tobacco powder, 2 kg of hot green chilli paste and 500 gram of garlic paste and stir it well with wooden stick in the clockwise direction, cover with gunny bag and leave for 24 hours under shade

Step 4: Next morning, add paste of any 10 types of leaves* (first five leaves is compulsory and choose 5 types of leaves from remaining list as per availability) to the mixture

Step 5: Stir thoroughly and cover with the gunny bag. Keep it for 30-40 days for fermentation so that the alkaloids present in the leaves will get dissolve in the mixture. Stir twice a day

Step 6: Filter this after 40 days with a muslin cloth and use it

How to Use?

2-3% Spray with water

Uses?

All types of sucking pest and for control of all caterpillars

Dashparni

Cow Urine



Sour butter milk

Materials required

- Water - 200 liters
- Fermented sour buttermilk - 6 liters

Procedure for preparation

- Take 6 litres of milk and make curds with it
- Remove the Creamy layer on it
- Let it remain for 3 to 5 days
- Churn it well and mix in water, filter and spray on infected trees

Dosage: 20 DAS(Days after sowing) -6 liters of sour buttermilk in 100 liters of water

Used: Works against all foliar diseases especially Powdery mildew, Downy mildew and Rust diseases



Sont astra/ Shunti astra

Materials required

- Water - 2 liters
- Ginger Powder - 200 grams
- Desi (native) cow milk - 2 liters

Procedure for preparation

- Take ginger powder and mix it in water
- Boil the solution till it becomes half by volume and let it cool
- Take Desi cowmilk in another bowl and boil it and remove the cream, Let it cool
- Mix both solutions and stir it well with a wooden pole, After 3 hours you can use it

Dosage: Obtained liquid to be diluted in 200 liters of water

Used against: management of plant diseases like Leaf spot



Some other pest control formulations

- Many farmers and NGOs have developed large number of innovative formulations which are effectively used for control of various pests.
- Although none of these formulations have been subjected to scientific validation but their wide acceptance by farmers speak of their usefulness.

Cow urine

- Cow urine, popularly known as "gomutra," ([germicidal](#), [antibiotic](#), [antimicrobial](#), and medicinal properties)
- [Nutrient-rich cow urine with nitrogen, potassium, and phosphorous](#) is highly beneficial to soil
- Besides macronutrients, the presence of [sulfur, sodium, manganese, iron, enzymes, and chlorine](#) make cow urine an integral natural pest repellent
- Cow urine diluted with water in **ratio of 1: 20** and used as foliar spray is not only effective in the management of pathogens & insects, but also acts as effective growth promoter for the crop



Some other pest control formulations

Broad spectrum formulation - 1

- In a copper container mix 3 kg fresh crushed neem leaves +1 kg neem seed kernel powder +10 liter of cow urine
- Seal the container and allow the suspension to ferment for 10 days
- After 10 days boil the suspension, till the volume is reduced to half
- Ground 500 gm green chilli in 1 liter of water and keep overnight
- In another container crush 250gm of garlic in water and keep overnight
- Next day mix and boil chilli extract and garlic extract
- Mix thoroughly all above and filter
- This is a broad spectrum pesticide and can be used on all crops against wide variety of insects
- Use 250 ml of this concentrate in 15 liter of water for spray



Some other pest control formulations

Broad spectrum formulation - 2

- Suspend **5 kg neem seed kernel powder** + **1kg Karanj seed powder** + **5 kg chopped leaves of Besharam** (*Ipomea* sp.) + **5kg chopped neem leaves**, Add all contents in one drum
- Add 10-12 liter of cow urine and fill the drum with water to make 150 liter
- Seal the drum and allow it to ferment for 8-10 days. After 8 days mix the contents and distil in a distiller
- Distillate will act as a good pesticide and growth promoter
- Distillate obtained from 150 liter liquid will be sufficient for one acre
- Dilute in appropriate proportion and use as foliar spray
- Distillate can be kept for few months without any loss in characteristics



Some other pest control formulations

Tutikadarasamis

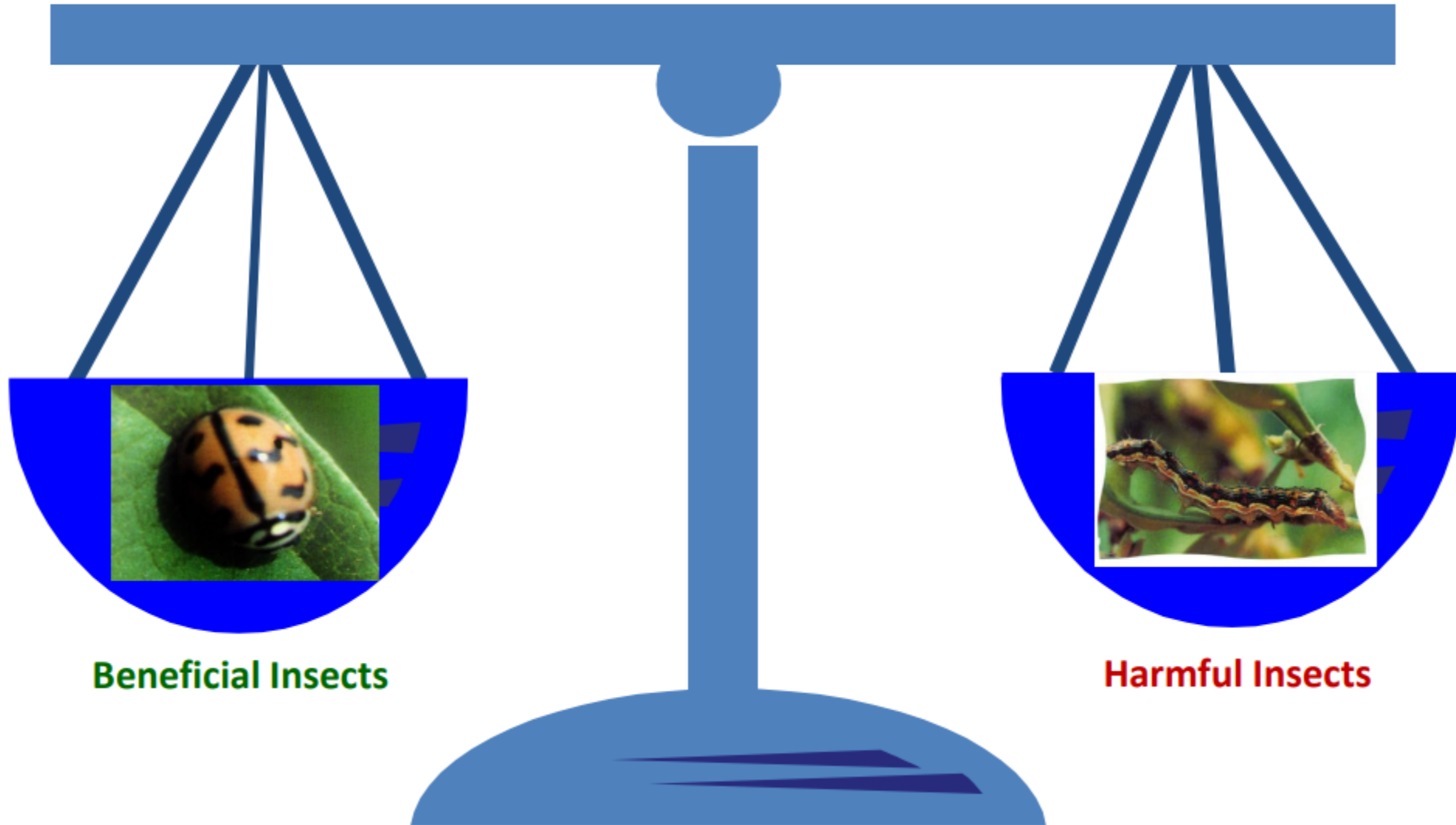
- Tutikadarasamis prepared from Datura leaves and cow urine
- The leaves are boiled in cow urine for 2-3 hours
- cooled and then filtered using cloth

Jungle Ki Kanddi

- Take kanddi powder (powder of indigenous cow's dung also known as jungle ki kanddi) and keep it in muslin cloth
- Tie one end of this bag in the centre of a wooden stick in such a way so that this bag hangs above the centre of the drum filled with 200 litres of water and leave for 48 hours
- Stir this solution two times in a day for 2-3 minutes in the clockwise direction
- The colour of the solution will change to a reddish brown colour (Katha/brass colour)
- After 48 hours, take out this bag and squeeze it, dip it again and then squeeze.
- Repeat this process three times
- Stir this solution properly. Spray this solution within 48 hours
- Before spraying, filter this solution



Balance of Nature





Thank you